Sharpness of RPCholesky error doubling with an explicit integer family

AI-assisted research project directed by Jitendra Prajapati

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Abstract

Chen, Epperly, Tropp, and Webber proved that the expected trace residual after exactly k randomly pivoted Cholesky pivots is at most 2^k times the optimal rank-k trace error. We give an explicit positive-definite family in dimension $k+1$ approaching this factor. A single rational parameter, explicit error bounds, and an integer rescaling yield finite witnesses with polynomial encoding length. On the same family, the normalized expected residual of size-k determinant sampling tends to $k+1$. The construction concerns exactly k pivots; one further pivot exhausts the family, so no oversampling conjecture is refuted. The proposed matching lower bound has undergone separately tasked AI checks, but human mathematical review, novelty, and publication priority remain unconfirmed.

1 Statement, algorithm, and attribution

For a real symmetric positive-definite $d \times d$ matrix A, ordinary randomly pivoted Cholesky (RPCholesky) starts with $R_0 = A$. Given R_j, it chooses i with probability $(R_j)_{ii} / \operatorname{tr} R_j$ and sets

$$R_{j+1} = R_j - \frac{R_j e_i e_i^{\mathsf{T}} R_j}{(R_j)_{ii}}.$$

An exhausted residual stays zero. Let $\tau_k(A) = \sum_{i>k} \lambda_i(A)$, with eigenvalues in decreasing order. Thus τ_k is the optimal rank-k trace error. All statements below use exact arithmetic and this prescribed pivot law. They concern an expectation over pivot choices, not a lower bound on every sampled path.

Theorem 1 (Sharpness of the published upper bound)**.** *For every integer $k \geq 1$,*

$$\sup_{d>k,\ A\succ 0} \frac{\mathbb{E} \operatorname{tr} R_k(A)}{\tau_k(A)} = 2^k. \tag{1}$$

The supremum is unchanged when A is restricted to real symmetric positive-definite integer matrices of dimension $k+1$.

The upper bound is the existing result of Chen, Epperly, Tropp, and Webber [1, Lemma 5.5]; it is not a contribution of this note. The proposed contribution is a matching explicit lower construction and its quantitative analysis. RPCholesky, fixed-size determinant sampling, Schur-complement identities, Cauchy–Binet, and cofactor identities are established ingredients. Perturbed-simplex lower bounds, including the $k = 1$ limiting factor 2, are established in [3, Proposition 3.3]. Constructions combining orthogonal simplices with separated scales are also established [4, Section 3]. The final inverse-diagonal and determinant-sampling identities used here are standard and are recorded, for example, in [5, Section 4.2]. No novelty is claimed for these ingredients, the weighted contrast basis, or the general idea of scale separation. The adaptive survivor-law analysis below concerns the specific ordinary RPCholesky sampling law. Theorem 1 is independent of any interpretation of a conjecture; the carefully qualified thesis implication appears in Section 9.

The proof will establish a rational one-parameter family with a uniform error estimate, then clear its denominators. This avoids a choice of successively smaller parameters without an explicit dependence between them. In particular, rare pivot paths are controlled after multiplication by their potentially large terminal amplification.

2 The explicit family and quantitative assertion

Fix $n = k + 1 \geq 2$ and $0 < t \leq 1/2$. Define

$$w_i = t^{i-1}, \qquad S_m = \sum_{i=1}^{m} w_i^2, \qquad Z = S_n, \qquad p_i = \frac{w_i^2}{Z}, \qquad h = t^{4n}, \qquad \delta = t^{4n(n-1)}. \tag{2}$$

For $1 \leq m < n$, let the weighted contrast $v_m \in \mathbb{R}^n$ have entries

$$(v_m)_i = \begin{cases} w_i w_{m+1}, & 1 \leq i \leq m, \\ -S_m, & i = m+1, \\ 0, & i > m+1. \end{cases} \qquad A_0 = \sum_{m=1}^{n-1} h^{m-1} v_m v_m^{\mathsf{T}}, \qquad A = A_0 + \delta I_n. \tag{3}$$

The arguments in Sections 3–5 first allow arbitrary $0 < h \leq 1$ and $\delta > 0$; the values in (2) are substituted only at the end.

Theorem 2 (Uniform finite-parameter estimate)**.** *The matrix A in (2)–(3) is positive definite, rational for rational t, and has $\tau_{n-1}(A) = \delta$. Moreover,*

$$\left| \frac{\mathbb{E}\,\mathrm{tr}\, R_{n-1}(A)}{\delta} - 2^{n-1} \right| \leq n 2^{n-1} t^2 + \left[2^n n(n-1) + 2(1 + (n-1)^2) \right] t^4 \tag{4}$$

$$\leq 2^{n+1} n^2 t^2. \tag{5}$$

Consequently, for $\eta > 0$, any integer

$$q \geq \max\{2, 2n/\sqrt{\eta}\}, \qquad t = 1/q, \tag{6}$$

gives a finite rational matrix satisfying

$$(1 - \eta) 2^{n-1} \leq \frac{\mathbb{E}\,\mathrm{tr}\, R_{n-1}(A)}{\tau_{n-1}(A)} \leq 2^{n-1}. \tag{7}$$

The last upper bound invokes [1, Lemma 5.5].

The constants and exponents are convenient safe choices; optimality of these choices is not asserted. For absolute error ϵ in the normalized expectation, it suffices to replace (6) by $q \geq \max\{2, \sqrt{2^{n+1} n^2/\epsilon}\}$.

3 Uniform spectral and principal-minor inputs

For $m < \ell$, the restriction of v_ℓ to the support of v_m is $w_{\ell+1} w$. Since $v_m^{\mathsf{T}} w = 0$, the vectors v_m are nonzero and mutually orthogonal. Directly, $\|v_m\|^2 = S_m S_{m+1}$. Hence A_0 has kernel $\mathrm{span}(w)$, and its nonzero eigenvalues are

$$\lambda_m = h^{m-1} S_m S_{m+1}, \qquad \lambda_* := \min_m \lambda_m \geq h^{n-2}. \tag{8}$$

No ordering of the individual λ_m is required. Adding δI makes the smallest eigenvalue exactly δ, proving the spectral-tail assertion of Theorem 2.

Write $u = w/\sqrt{Z}$ and $p_{\min} = p_n$. The semidefinite inequality $A_0 \succeq \lambda_*(I - uu^{\mathsf{T}})$ gives, for every nonempty proper coordinate set T,

$$A_0[T, T] \succeq h^{n-2} p_{\min} I_T =: \mu I_T. \tag{9}$$

Indeed the least eigenvalue of $I_T - u_T u_T^{\mathsf{T}}$ is $1 - \|u_T\|^2 = \sum_{j \notin T} p_j \geq p_{\min}$. Thus every proper principal submatrix is positive definite. Also

$$1 \leq Z \leq \frac{4}{3}, \qquad p_{\min}^{-1} \leq 2t^{-2(n-1)}, \qquad p_{\min}^{-2} \leq 2t^{-4(n-1)}. \tag{10}$$

The last constant follows from $Z^2 \leq 16/9 < 2$, independently of the preceding, looser factor-2 bound.

Let V have rows $v_1, \dots, v_{n-1}$. For $r \le n-1$, restrict its first r rows to the first $r+1$ coordinates, obtaining H_r. Its kernel is spanned by $(w_1, \dots, w_{r+1})$, so its signed maximal cofactors are proportional to these entries. Cauchy–Binet determines the constant:

$$\det(H_r \text{ with column } j \text{ omitted})^2 = w_j^2 \left(\prod_{\ell=1}^{r} S_\ell \right)^2. \tag{11}$$

For completeness, the sum of the squared cofactors is $\det(H_r H_r^{\mathsf{T}}) = \prod_{\ell=1}^{r} S_\ell S_{\ell+1}$. Divide by $\sum_{j=1}^{r+1} w_j^2 = S_{r+1}$ and use $S_1 = 1$ to obtain (11).

An additional coefficient-sum bound will avoid any hidden combinatorial factor. Put $B = \sum_{m=1}^{n-1} v_m v_m^{\mathsf{T}}$. Its nonzero eigenvalues are $S_m S_{m+1}$, so $\|B\| \le 16/9 < 2$. For every size-r coordinate set T,

$$\sum_{\substack{I \subseteq \{1,\dots,n-1\} \\ |I|=r}} \det(V[I,T])^2 = \det B[T,T] \le 2^r. \tag{12}$$

This bounds the sum of all squared-minor coefficients, not just each coefficient separately.

4 The hierarchical law and its weighted error

After selecting a coordinate set S, the residual diagonal at unused i equals $\det A_0[S \cup \{i\}, S \cup \{i\}] / \det A_0[S,S]$. Consequently the next-pivot law on A_0 is

$$P_0(i \mid S) = \frac{\det A_0[S \cup \{i\}, S \cup \{i\}]}{\sum_{j \notin S} \det A_0[S \cup \{j\}, S \cup \{j\}]}. \tag{13}$$

This is a transition identity; it does not identify the final RP subset law with determinant sampling. All blocks used have size at most $n-1$.

Call a size-m selected set *legal* if $S = \{1, \dots, m+1\} \setminus \{J\}$ for some $J \le m+1$. The empty set is legal with representative $J = 1$. At a legal state with $m \le n-2$, put $r = m+1$ and $e = r(r-1)/2$. Cauchy–Binet gives

$$\det A_0[T,T] = \sum_{|I|=r} h^{\sum_{a \in I}(a-1)} \det(V[I,T])^2, \qquad |T| = r. \tag{14}$$

The unique least exponent is e, from $I = \{1, \dots, r\}$; every other exponent is at least $e+1$. For $T = S \cup \{i\}$, the leading coefficient is nonzero only for $i = J$ or $i = m+2$. By (11), these coefficients are

$$c_J = w_{m+2}^2 \left(\prod_{\ell=1}^{r} S_\ell \right)^2, \qquad c_{m+2} = w_J^2 \left(\prod_{\ell=1}^{r} S_\ell \right)^2. \tag{15}$$

For other unused i, set $c_i = 0$. Write each determinant weight as $h^e c_i + E_i$. Nonnegative coefficients, $h \le 1$, and (12) imply $0 \le E_i \le 2^r h^{e+1}$. Thus the total leading and remainder weights obey

$$L := h^e \sum_i c_i \ge h^e t^{2(n-1)}, \qquad E := \sum_i E_i \le n 2^{n-1} h^{e+1}. \tag{16}$$

The lower bound is deliberately loose: $J \le n-1$ and each $S_\ell \ge 1$.

Let Q normalize the two leading coefficients. The actual transition is exactly the mixture $(L/(L+E))Q + (E/(L+E))U$, where U normalizes the remainders when $E > 0$. Its total variation distance from Q is at most

$$\alpha := n 2^{n-1} h\, t^{-2(n-1)}. \tag{17}$$

We use d_{TV} as the maximal difference of event probabilities, equal to the smallest possible disagreement probability in a coupling. Couple the RP path and the Q path while they agree. The latter always stays legal; a union bound over $n-1$ transitions bounds the probability of any disagreement by $(n-1)\alpha$. This includes all illegal RP paths, and requires no estimate after the first disagreement.

Let J now denote the final unselected coordinate and put $f(J) = 1/p_J$. Because $0 < f \le 1/p_{\min}$, (10) gives the crucial weighted bound

$$|\mathbb{E}_{P_0} f(J) - \mathbb{E}_Q f(J)| \le 2^n n(n-1) h\, t^{-4(n-1)}. \tag{18}$$

In particular, convergence of transition probabilities alone has not been used to justify convergence of this unbounded-as-$t \to 0$ payoff.

The law Q is an explicit tournament. Starting with $J_1 = 1$, introduce $j = 2, \dots, n$; if the old representative is $i < j$, it survives with probability $p_i/(p_i + p_j)$, and the new representative is j otherwise. The opposite-column coefficients in (15) give exactly these survival probabilities. If $F_j = \mathbb{E}[1/p_{J_j}]$, then

$$\mathbb{E}[1/p_{J_j} \mid J_{j-1} = i] = \frac{2}{p_i + p_j}, \qquad F_1 = Z.$$

Since $p_j/p_i = t^{2(j-i)} \le t^2$,

$$Z\left(\frac{2}{1+t^2}\right)^{n-1} \le F_n \le Z 2^{n-1}, \qquad |F_n - 2^{n-1}| \le n 2^{n-1} t^2. \tag{19}$$

For the last inequality use $(1+x)^{-(n-1)} \ge 1 - (n-1)x$ and $(1-t^2)^{-1} \le 1 + \frac{4}{3}t^2$ for $t \le 1/2$.

5 Uniform noise control and completion of the bound

Let P_δ be the RP path law for $A_0 + \delta I$. For a proper set T of size r, (9) implies, with $x = \delta/\mu$,

$$1 \le \frac{\det(A_0[T,T] + \delta I_T)}{\det A_0[T,T]} \le (1+x)^r.$$

Every transition weight in (13) is therefore tilted by a factor between 1 and $\beta = (1+x)^{n-1}$. It follows pointwise that $P_\delta(i \mid S) \ge P_0(i \mid S)/\beta$. The transition-law overlap is at least β^{-1}, so its total variation distance is at most $1 - \beta^{-1} \le (n-1)x$. This last inequality holds for every $x \ge 0$. Coupling all $n-1$ transitions, now at arbitrary selected states, gives

$$d_{\mathrm{TV}}(P_\delta, P_0) \le (n-1)^2 \frac{\delta}{\mu}. \tag{20}$$

After $n-1$ pivots, the scalar Schur-complement identity gives $\operatorname{tr} R_{n-1}(A) = 1/(A^{-1})_{JJ}$. Spectral decomposition yields

$$(A^{-1})_{jj} = p_j/\delta + b_j, \qquad 0 \le b_j \le \lambda_*^{-1} \le h^{-(n-2)}.$$

Thus the normalized endpoint residual is $g_\delta(j) = 1/(p_j + \delta b_j)$, and

$$0 \le f(j) - g_\delta(j) \le \frac{\delta h^{-(n-2)}}{p_{\min}^2}. \tag{21}$$

Comparing $\mathbb{E}_{P_\delta} g_\delta$ first with $\mathbb{E}_{P_\delta} f$ and then with $\mathbb{E}_{P_0} f$, equations (20)–(21) and $\mu = h^{n-2} p_{\min}$ show

$$\begin{aligned} |\mathbb{E}_{P_\delta} g_\delta(J) - \mathbb{E}_{P_0} f(J)| &\le [1 + (n-1)^2] \frac{\delta h^{-(n-2)}}{p_{\min}^2} \\ &\le 2[1 + (n-1)^2] \delta h^{-(n-2)} t^{-4(n-1)}. \end{aligned} \tag{22}$$

The comparison payoff f under P_0 is a limiting amplification weight; it is not the residual trace of the rank-$(n-1)$ matrix A_0, which vanishes after $n-1$ pivots. This distinction accounts for both endpoint and path-law perturbations.

Completion of Theorem 2. Substitute $h = t^{4n}$ and $\delta = t^{4n(n-1)}$. The two dangerous weighted factors become

$$h t^{-4(n-1)} = t^4, \qquad \delta h^{-(n-2)} t^{-4(n-1)} = t^4. \tag{23}$$

The triangle inequality applied to (18), (19), and (22) proves (4). Since $t^4 \le t^2$ and

$$n 2^{n-1} + 2^n n(n-1) \le 2^n n^2, \qquad 2[1 + (n-1)^2] \le 2n^2 \le 2^n n^2,$$

we obtain (5). Dividing by 2^{n-1} and taking $t = 1/q$ gives relative error at most $4n^2/q^2$, proving the tolerance claim. The published upper bound supplies the upper side of (7). □

6 Integer matrices and encoding length

The rational family admits a formula using only integer powers and sums. Let $q \geq 2$ be an integer, $t = 1/q$, and $D = 4n(n-1)$. Define

$$(z_m)_i = \begin{cases} q^{m-i}, & 1 \leq i \leq m, \\ -\sum_{\ell=0}^{m-1} q^{2m-1-2\ell}, & i = m+1, \\ 0, & i > m+1. \end{cases} \tag{24}$$

These are precisely $z_m = q^{2m-1} v_m$. Therefore

$$M_{n,q} := I_n + \sum_{m=1}^{n-1} q^{D-4n(m-1)-4m+2} z_m z_m^{\mathsf{T}} = q^D A. \tag{25}$$

The coefficient exponent decreases with m and equals 6 at $m = n-1$; all coefficients are positive integers. The z_m are orthogonal and orthogonal to $(q^{n-1}, q^{n-2}, \ldots, q, 1)$. Hence $M_{n,q}$ is integer SPD, its least eigenvalue is 1, and $\tau_{n-1}(M_{n,q}) = 1$.

Multiplication by a positive scalar leaves every RP pivot probability unchanged and scales every residual and spectral tail by that scalar. Theorem 2 therefore gives the following consequence.

Corollary 1 (Finite integer witness)**.** *For $n \geq 2$, $\eta > 0$, and an integer q satisfying (6), the explicit integer matrix (25) satisfies*

$$\tau_{n-1}(M_{n,q}) = 1, \qquad (1-\eta)2^{n-1} \leq \mathbb{E}\,\mathrm{tr}\, R_{n-1}(M_{n,q}) \leq 2^{n-1}.$$

Each entry has $O(n^2 \log q)$ bits, and the full dense matrix has $O(n^4 \log q)$ bits. With $0 < \eta < 1$ and $q = \lceil 2n/\sqrt{\eta} \rceil$, this is $O(n^4(\log n + \log(1/\eta)))$ bits in total.

Proof. Only the size bound remains. Since $h \leq 1$, $A_0 \preceq B$, and (2) gives $\delta \leq 1$. Thus $\|A\| \leq 16/9 + 1 < 3$. Every entry of $M_{n,q} = q^D A$ has absolute value less than $3q^D$, proving the per-entry and dense encoding bounds. The stated choice of q has $\log q = O(\log n + \log(1/\eta))$. □

This is a polynomial encoding-length statement, not a polynomial-time claim for exact expectation evaluation. The small finite checks below enumerate pivot paths. No numerical conditioning or floating-point stability guarantee follows from a short exact encoding.

Proof of Theorem 1. The published upper bound applies to every SPD input. For fixed k, put $n = k+1$ in Corollary 1, and let $\eta \downarrow 0$. Every chosen tolerance gives a finite integer SPD matrix, so the supremum over that restricted class is at least 2^k. Together with the upper bound this proves both assertions. No singular limiting matrix is used as an input, and no finite attainment of the supremum is asserted. □

7 Same-size determinant sampling

A size-k determinant sample (a k-DPP with kernel A) selects T with probability $\det A[T,T]/e_k(A)$, where e_k is the kth elementary symmetric polynomial of the eigenvalues. Its residual is the same selected-set Schur residual. For $k = n-1$, every set omits one j. Multiplying its weight by its residual cancels its principal determinant, giving the exact identity

$$\mathbb{E}\,\mathrm{tr}\, R_{n-1}^{\mathrm{DPP}}(A) = \frac{n \det A}{e_{n-1}(A)} = \frac{n}{\mathrm{tr}(A^{-1})}. \tag{26}$$

By (8),

$$\frac{\mathbb{E}\,\mathrm{tr}\, R_{n-1}^{\mathrm{DPP}}(A)}{\delta} = \frac{n}{1 + \delta \sum_{m=1}^{n-1} (\lambda_m + \delta)^{-1}}, \qquad 0 \leq \delta \sum_m (\lambda_m + \delta)^{-1} \leq (n-1)t^{4n}.$$

Consequently

$$\frac{n}{1 + (n-1)t^{4n}} \leq \frac{\mathbb{E}\,\mathrm{tr}\, R_{n-1}^{\mathrm{DPP}}(A)}{\delta} \leq n, \qquad \lim_{t \downarrow 0} \frac{\mathbb{E}\,\mathrm{tr}\, R_{n-1}^{\mathrm{RP}}(A)}{\mathbb{E}\,\mathrm{tr}\, R_{n-1}^{\mathrm{DPP}}(A)} = \frac{2^{n-1}}{n}. \tag{27}$$

For $0 < \eta < 1$, the integer tolerance choice gives a finite RP/DPP expected-error ratio at least $(1-\eta)2^{n-1}/n$. Thus no universal constant or polynomial in k bounds the same-k ratio for all SPD matrices. The DPP value $k+1$ is a family limit, not an equality asserted for every finite member. This compares trace errors at the same number of selected coordinates, not runtimes or practical kernel performance.

8 Illustrative exact finite checks

The accompanying integer-family records contain ten identity cases ($n = 2, \dots, 6$, $q = 2, 3$) and four complete exact expectation cases, all recorded as passing the fixed protocol. The identity checks compare (25) entry by entry with $q^D A$ and check its full orthogonal eigenbasis. For the expectation cases, the smallest admissible integer q in (6) was chosen with $\eta = 1/10$. The recorded values are:

k	q	$\mathbb{E}\,\mathrm{tr}\,R_k(M_{k+1,q})$	Ratio to 2^k	Max. entry bits
1	13	1.9999999976	0.9999999988	30
2	19	3.9944903578	0.9986225894	102
3	26	7.9881918819	0.9985239852	226
4	32	15.9765967765	0.9985372985	401

Here $\tau_k = 1$ exactly. Displayed decimals are rounded; all pass/fail comparisons use rational arithmetic. The direct recursion weights every positive residual diagonal by its fraction of the trace, performs the Schur update, and enumerates the remaining paths. A known 3×3 control gives $23/28$; an intentionally uniform-pivot mutation gives a different result and is rejected. The saved finite expectations are strictly below 2^k.

These checks illustrate the construction and test the implementation. They are not Monte Carlo estimates, benchmarks, or evidence by finite extrapolation for arbitrary n. The theorem and its tolerance guarantee rest on the proof above. A separate AI audit checked code, saved-record consistency, and hashes without independently rerunning the expectation recursion; it should not be described as an additional execution.

9 Scope, thesis interpretation, and review status

The sharpness theorem rules out a universal polynomial $C(r)$ satisfying

$$\mathbb{E}\,\mathrm{tr}\,R_r(A) \le C(r)\tau_r(A) \quad \text{for all positive-definite } A. \tag{28}$$

Indeed choose r so that $2^r > C(r)$, then choose the finite tolerance small enough to violate the proposed inequality. This conclusion is independent of any ambiguity in an external statement.

Epperly's thesis [2, Conjecture 11.2, printed p. 182] proposes an improved polynomial r-step bound. The printed display is

$$\mathrm{tr}(A - A^{(r)}) \le C(r)\,\mathrm{tr}(A - [A]_r), \qquad C(r) = \mathrm{poly}(r).$$

There are two notation issues: the thesis defines $A^{(r)}$ as a residual, making the displayed subtraction an approximation trace, and the display omits expectation. Its adjacent paragraph cites CETW25 Lemma 5.5 as establishing the relevant bound with $C(r) = 2^r$. That citation and the surrounding discussion support the expected-residual interpretation (28). *Under that interpretation*, Theorem 1 contradicts the proposed polynomial bound. This repaired expression is an interpretation, not a verbatim quotation of the printed equation, and the author's intended notation has not been confirmed. The project source audit checked both the official deposit and arXiv:2512.15929v1 of 17 December 2025. The primary claim of this note remains sharpness of the independently stated published upper bound.

Every matrix constructed here has dimension $r + 1$ and is exhausted by its next pivot. The argument therefore does not refute the thesis's separate oversampling conjecture, the $O(r/\varepsilon)$ question discussed in [6], or valid oversampling guarantees. It also does not establish a median-error or fixed-confidence obstruction. Rare paths can matter to the expectation, and no practical application degradation is claimed.

Attribution and status. The project was directed by Jitendra Prajapati. OpenAI Codex produced the construction, derivation, explicit integer rescaling, implementation, and source-audit work. Separately tasked AI agents checked the main mechanism, uniform perturbation bounds, exponent arithmetic, integer rescaling,

finite-control artifacts, and source scope. These checks are not formal verification or independent human mathematical validation. Bounded literature searches did not establish priority; novelty and publication priority remain unconfirmed. This public research draft has not been peer reviewed and claims no external endorsement or journal acceptance. Independent human mathematical review remains pending.

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